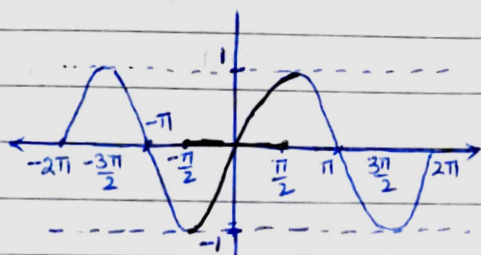


Key Topics:

1. Inverse Trigonometric functions (ITF)
2. Properties of ITF.

Inverse Sine function:

- $\sin: \mathbb{R} \rightarrow \mathbb{R}$ $\begin{cases} \rightarrow \text{not one-one} \\ \rightarrow \text{not onto} \end{cases}$
- We restrict the domain to $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, so that the $\sin x$ becomes one-one.
- We restrict the co-domain to $[-1, 1]$, so that $\sin x$ becomes onto.



Thus $\sin: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1]$
is bijective & hence
invertible.

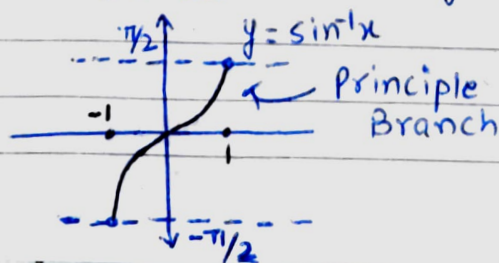
So, $\sin^{-1}: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ Principle Range

Definition: $y = \sin x \Leftrightarrow x = \sin^{-1} y$

where $y \in [-1, 1]$ or $|y| \leq 1$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ or $|x| \leq \frac{\pi}{2}$

Properties of Inverse Sine function:

- Domain of $\sin^{-1} x$ is $[-1, 1]$ and range is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- $\sin(\sin^{-1} x) = x$, where $x \in [-1, 1]$
- $\sin^{-1}(\sin x) = x$ where $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- $\sin^{-1}: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is strictly increasing.

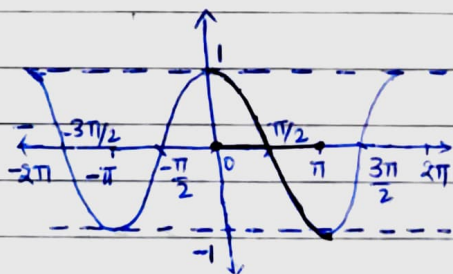




★ Inverse Cosine function:

- $\cos: \mathbb{R} \rightarrow \mathbb{R}$ $\left\{ \begin{array}{l} \rightarrow \text{not one-one} \\ \rightarrow \text{Not onto} \end{array} \right.$
- We restrict domain to $[0, \pi]$, so that $\cos x$ is one-one.
- We restrict co-domain to $[-1, 1]$, so that $\cos x$ is onto.
- Hence $\cos: [0, \pi] \rightarrow [-1, 1]$ is bijective, so it's invertible.

$$\Rightarrow \cos^{-1}: [-1, 1] \rightarrow [0, \pi] \quad \leftarrow \text{Principle Range}$$



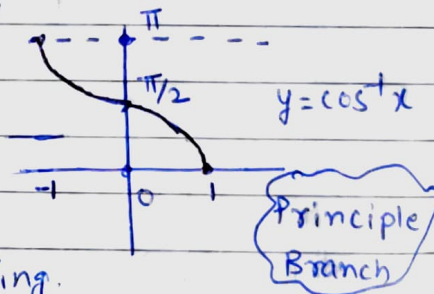
Definition:

$$y = \cos x \Leftrightarrow x = \cos^{-1} y$$

where $y \in [-1, 1]$
and $x \in [0, \pi]$

• Properties of Inverse Cosine function:

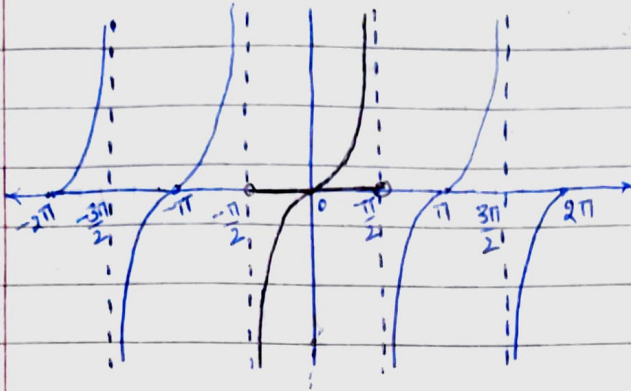
- Domain of $\cos^{-1} x$ is $[-1, 1]$ and range is $[0, \pi]$
- $\cos(\cos^{-1} x) = x$, where $x \in [-1, 1]$
- $\cos^{-1}(\cos x) = x$, where $x \in [0, \pi]$
- $\cos^{-1}: [-1, 1] \rightarrow [0, \pi]$ is strictly decreasing.



★ Inverse Tangent function:

- $\tan: A \rightarrow \mathbb{R}$, where $A = \mathbb{R} - \{(2n+1)\frac{\pi}{2} : n \in \mathbb{Z}\}$ $\left\{ \begin{array}{l} \rightarrow \text{not one-one} \\ \rightarrow \text{not onto} \end{array} \right.$
- We restrict domain $(-\frac{\pi}{2}, \frac{\pi}{2})$, so that $\tan x$ is one-one.
- hence, $\tan x: (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow \mathbb{R}$ is bijective.
- $\tan: (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow \mathbb{R}$ is invertible.

$$\text{So, } \tan^{-1}: \mathbb{R} \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2}) \quad \leftarrow \text{Principle Range}$$

Definition:

$$y = \tan x \Leftrightarrow x = \tan^{-1} y$$

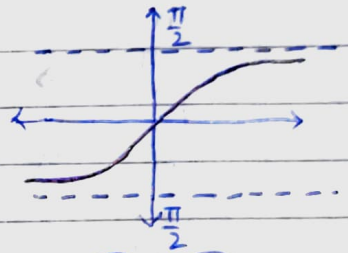
where $y \in \mathbb{R}$

$$\text{and } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\text{i.e. } |x| < \frac{\pi}{2}$$

Properties of Inverse Tangent function:

- Domain of $\tan^{-1} x$ is $\mathbb{R}$ and range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
- $\tan(\tan^{-1} x) = x$, where $x \in \mathbb{R}$
- $\tan^{-1}(\tan x) = x$, where $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
- $\tan^{-1}: \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ is strictly increasing



$$y = \tan^{-1} x$$

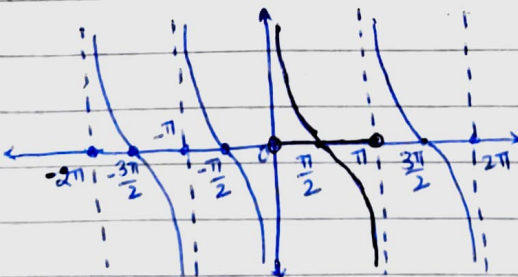
Principle Branch

★ Inverse Cotangent function:

- $\cot: A \rightarrow \mathbb{R}$, where $A = \mathbb{R} - \{n\pi : n \in \mathbb{Z}\}$ $\rightarrow$ not one-one
 $\rightarrow$ not onto

- We restrict the domain to $(0, \pi)$, so that $\cot: (0, \pi) \rightarrow \mathbb{R}$ is ~~that~~ one-one (onto), thus it is bijective, hence it is invertible.

$$\cot^{-1}: \mathbb{R} \rightarrow (0, \pi)$$

Definition:

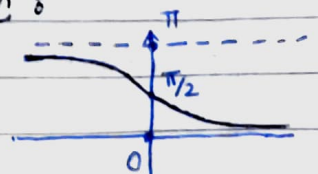
$$y = \cot x \Leftrightarrow x = \cot^{-1} y$$

where $y \in \mathbb{R}$

$$\text{and } x \in (0, \pi)$$

Properties of Inverse Cotangent function:

- Domain of $\cot^{-1} x$ is $\mathbb{R}$ and range is $(0, \pi)$
- $\cot(\cot^{-1} x) = x$ for $x \in \mathbb{R}$
- $\cot^{-1}(\cot x) = x$ for $x \in (0, \pi)$
- $\cot^{-1}: \mathbb{R} \rightarrow (0, \pi)$ is strictly decreasing.



$$y = \cot^{-1} x$$

Principle Branch



★ Inverse Secant function:

• $\sec: A \rightarrow R$

where $A = R - \{(2n+1)\frac{\pi}{2} : n \in \mathbb{Z}\}$

→ not one-one
→ not onto

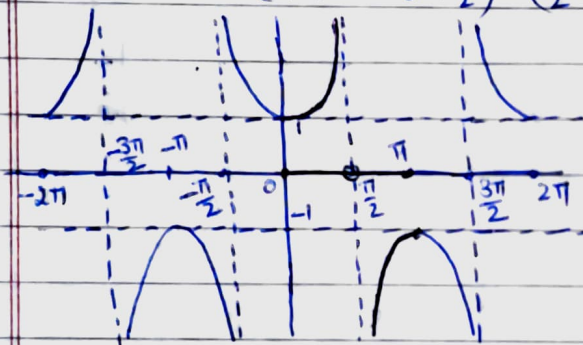
• We restrict the domain to $[0, \pi] - \{\frac{\pi}{2}\} = [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$, so that $\sec x$ is one-one.

• If we restrict R (co-domain) to $(-\infty, -1] \cup [1, \infty)$, so that $\sec x$ is onto

• Hence, $\sec: [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi] \rightarrow R - (-1, 1)$ is bijective.

• Hence, it is invertible.

• $\sec^{-1}: R - (-1, 1) \rightarrow [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$



Definition:

$$y = \sec x \Leftrightarrow x = \sec^{-1} y$$

where $y \in (-\infty, -1] \cup [1, \infty)$

and $x \in [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$

• Properties of Inverse Secant function:

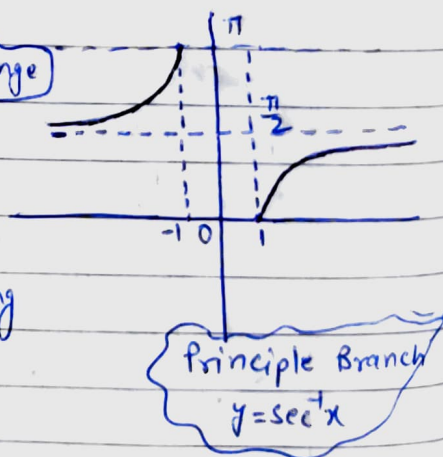
• Domain of $\sec^{-1} x$ is $(-\infty, -1] \cup [1, \infty)$

and range is $[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$ ← Principle Range

• $\sec(\sec^{-1} x) = x$ for $x \in (-\infty, -1] \cup [1, \infty)$

• $\sec^{-1}(\sec x) = x$ for $x \in [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$

• $\sec^{-1} x: (-\infty, -1] \cup [1, \infty) \rightarrow [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$ is strictly increasing



★ Inverse Cosecant function:

• $\operatorname{cosec}: A \rightarrow R$, where $A = R - \{n\pi : n \in \mathbb{Z}\}$

→ Not one-one
→ Not onto

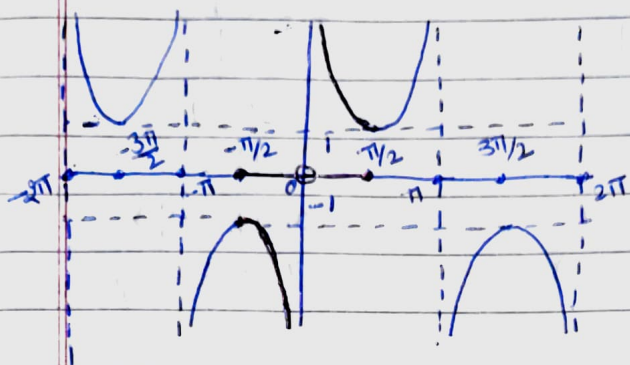
• So, we restrict the domain to $[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\} = [-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$ and co-domain to $(-\infty, -1] \cup [1, \infty) = R - (-1, 1)$ to make it bijective function.

~~single~~



So, $\operatorname{cosec} : \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right] \rightarrow (-\infty, -1] \cup [1, \infty)$ is invertible.

So, $\operatorname{cosec}^{-1} : (-\infty, -1] \cup [1, \infty) \rightarrow \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$



Definition:

$y = \operatorname{cosec} x \Leftrightarrow x = \operatorname{cosec}^{-1} y$
where $y \in (-\infty, -1] \cup [1, \infty)$
and $x \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$

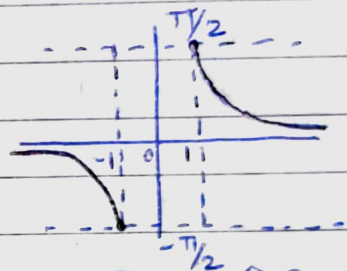
• Properties of Inverse Cosecant function:

• Domain of $\operatorname{cosec}^{-1} x$ is $(-\infty, -1] \cup [1, \infty)$
and range is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

• $\operatorname{cosec}^{-1}(\operatorname{cosec} x) = x$ for $x \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$

• $\operatorname{cosec}(\operatorname{cosec}^{-1} x) = x$ for $x \in (-\infty, -1] \cup [1, \infty)$

• $\operatorname{cosec}^{-1} : (-\infty, -1] \cup [1, \infty) \rightarrow \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$
is strictly decreasing



$y = \operatorname{cosec}^{-1} x$
Principle Branch

Theorems:

- | | |
|--|---|
| ★ 1. $\sin^{-1}(-x) = -\sin^{-1} x$ | 4. $\cos^{-1}(-x) = \pi - \cos^{-1} x$ $\forall x \leq 1$ |
| 2. $\tan^{-1}(-x) = -\tan^{-1} x$ | 5. $\cot^{-1}(-x) = \pi - \cot^{-1} x$ $\forall x \in \mathbb{R}$ |
| 3. $\sec^{-1}(-x) = \pi - \sec^{-1} x$ | 6. $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}(x)$ $\forall x \geq 1$ |

★ 1. $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$

2. $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$

3. $\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}$

★ 1. $\operatorname{cosec}^{-1}(x) = \sin^{-1}\left(\frac{1}{x}\right)$

2. $\sec^{-1}(x) = \cos^{-1}\left(\frac{1}{x}\right)$

3. $\cot^{-1}(x) = \begin{cases} \tan^{-1}\left(\frac{1}{x}\right) & ; x \geq 0 \\ \pi + \tan^{-1}\left(\frac{1}{x}\right) & ; x < 0 \end{cases}$

★ Sum up :

Function	Domain	Range
$\sin^{-1}x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos^{-1}x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1}x$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$\cot^{-1}x$	$\mathbb{R}$	$(0, \pi)$
$\sec^{-1}x$	$\mathbb{R} - (-1, 1)$	$[0, \pi] - \{\frac{\pi}{2}\}$
$\csc^{-1}x$	$\mathbb{R} - (-1, 1)$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$

Recall some useful Trigonometric Identities :

- $\sin 3x = 3\sin x - 4\sin^3 x$
- $\cos 3x = 4\cos^3 x - 3\cos x$
- $\tan 3x = \frac{3\tan x - \tan^3 x}{1 - 3\tan^2 x}$
- $\sin 2x = 2\sin x \cos x$
- $\cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1 = \frac{1 - \tan^2 x}{1 + \tan^2 x}$
- $\tan 2x = \frac{2\tan x}{1 - \tan^2 x}$
- $\sin 2x = \frac{2\tan x}{1 + \tan^2 x}$

★ Now using in the I.T.F :-

- $3\sin^{-1}x = \sin^{-1}(3x - 4x^3), x \in [-\frac{1}{2}, \frac{1}{2}]$
- $3\cos^{-1}x = \cos^{-1}(4x^3 - 3x), x \in [\frac{1}{2}, 1]$
- $2\tan^{-1}x = \tan^{-1}\left(\frac{2x}{1-x^2}\right); |x| \leq 1$
- $2\tan^{-1}x = \sin^{-1}\left(\frac{2x}{1+x^2}\right); |x| \leq 1$
- $2\tan^{-1}x = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right); x \geq 0$
- $3\tan^{-1}x = \tan^{-1}\left(\frac{3x - x^3}{1 - 3x^2}\right); x \in [-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}]$
- $2\sin^{-1}x = \sin^{-1}(2x\sqrt{1-x^2}); |x| \leq 1$